Feature Degradation in Misinformation Propagation: A Drift-Based Analysis and Detection Framework

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Abstract

Misinformation propagation leaves a measurable, but modest, trace in how content features change along a propagation tree. We define *drift* as the parent–child change in node representations and find that fake-labelled cascades drift less at matched depth on GossipCop (2.3%; hop-standardised ratio 0.977; cascade-level $p = 0.048$); the corresponding PolitiFact cascade-level comparison is not significant ($p = 0.52$). Content-drift statistics add information beyond cascade size in an internal cross-validated analysis (+0.054 AUC; paired bootstrap $p < 10^{-4}$). Fake cascades also contain more near-duplicate siblings, and sibling duplication is associated with lower drift within each class and after controlling for cascade size. These observational results are consistent with a copying-based explanation but do not establish coordination or causality. Per-node drift features do not improve the classifier, and the signal weakens in partial cascades. The contribution is a controlled measurement, mechanism-consistent evidence, and a design lesson—not a state-of-the-art detector.

1 Introduction

Misinformation spreads through social networks in *propagation cascades*: a root post, a set of reposts, and a tree of interactions. Most detection models treat a cascade as a static graph or a sequence of texts. We ask whether cascades carry an additional signal: **content features change as they propagate**. One hypothesis is that real-news cascades accumulate more heterogeneous commentary, while some fabricated-news cascades contain more repeated content. If so, this difference may leave a measurable trace in the *feature drift* between a node and its parent. We test that account as an empirical hypothesis, not as an assumption about every cascade.

This work makes the following contributions:

1. **A cross-dataset empirical finding, with its confound removed.** Fake news propagates with *less* feature drift than real news. Pooling all test nodes gives a large ratio (0.901 on GossipCop, $t = -63.6$), but that statistic mixes the content effect with a structural one: real cascades are hop-1-heavy and hop-1 nodes carry far more drift than deep nodes. Controlling for propagation depth, the effect is **∼2.3% on GossipCop** (hop-standardised ratio 0.977; cascade-level ratio 0.975, $p = 0.048$) and **not significant at cascade level on PolitiFact** (ratio 0.969, $p = 0.52$). We report the confounded and the controlled statistics side by side.

2. **Evidence that the signal is not a proxy for cascade shape.** Cascade size alone reaches 0.792 AUC on GossipCop, so we test what drift adds: content drift beats size alone by +0.054 **AUC** (paired bootstrap over identical folds, $p < 10^{-4}$), beats structure at matched size (+0.015, $p = 0.004$), and adds more than structure does when combined with size. The information is about the *content* relationship between a node and its parent, not the tree's geometry.

3. **Evidence consistent with a copying-based explanation.** We measure the **sibling near-duplicate rate**—the fraction of nodes sharing a parent whose content is near-identical—and find fake-news cascades contain more near-duplicate siblings (98.3% vs. 92.7% above cosine 0.95, $p = 10^{-287}$), while *overall* within-cascade dispersion does not differ. Near-duplication is negatively associated with drift ($r = -0.31$, $p = 2 \times 10^{-85}$), within each class and after controlling for cascade size, and discriminates on its own at 0.831 AUC. This pattern supports a copying-based account, but these observational embedding analyses do not identify coordinated actors or establish a causal mechanism.

4. **A design rule, from an honest negative result.** Per-node drift features *hurt* the classifier at every cascade size, while graph-level drift statistics help. This reconciles the discriminator result with

the ablation and yields a concrete rule: consume drift via set-pooling, not concatenation.

5. **The best detector in our comparison, and what it is worth. HybridDegradLSTM** reaches **95.85% accuracy / 0.9876 AUC** on GossipCop—the best of the six models we compare against, verified over 3 seeds. We do not claim it beats published UPFD results, and its margin over a strong LSTM baseline is not statistically significant. We report it as an engineering artifact illustrating the design rule, not as the paper's headline. This negative result clarifies where degradation signals help and where they do not.

2 Background and Related Work

Fake news detection on graphs. The UPFD benchmark [1] provides tree-structured cascades from PolitiFact and GossipCop with multiple feature sets. Standard graph models include GCN [3], GAT [4], and BiGCN [2], which performs top-down and bottom-up message passing over the propagation tree.

Sequential models for rumors. Propagation can be read as a sequence: **(author?)** [5] and **(author?)** [6] use recurrent and convolutional networks over propagation paths. Our BaselineLSTM orders nodes by propagation depth and encodes them with a bidirectional LSTM.

Content-based models. Pre-trained text encoders (e.g., BERT [7]) provide per-post representations; a mean-pooling MLP (BERT-MLP) is a common text-only baseline.

Degradation / drift in cascades. Prior rumor-detection work has modeled "temporal" and "structural" signals but, to our knowledge, has not quantified the *content drift between parent and child nodes* as a first-class phenomenon. Our drift features formalize this: for each node, the distance (cosine and ℓ_2) of its feature vector from its parent's, together with structural context (depth, number of children, rootness).

3 Method

3.1 Drift features

Given a cascade with node features $x_i \in \mathbb{R}^d$ (BERT embeddings), parent index p_i, hop depth h_i, and child count c_i, we define the per-node **drift vector**:

$$d_i = \Big[1-\cos(x_i, x_{p_i}),\ \|x_i - x_{p_i}\|_2,\ \tfrac{h_i}{\max h},\ \tfrac{c_i}{\max c},\ \mathbb{1}[h_i = 0]\Big] \in \mathbb{R}^5, \tag{1}$$

where the two distance terms are zeroed for the root (masked by parent existence). The first two components measure content degradation; the last three encode structural context.

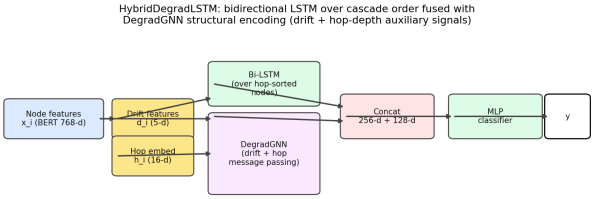


Figure 1: HybridDegradLSTM fuses a bidirectional LSTM over cascade order with the DegradGNN structural encoding (drift + hop-depth auxiliary signals).

3.2 DegradGNN

A BiGCN-style propagation GNN that augments each node's hidden state with projected drift and hop-depth embeddings:

$$h_i^{(0)} = W_{\text{proj}}\, x_i, \qquad g_i = \big[\, W_{\text{hop}}\, e_{h_i},\ W_{\text{drift}}\, d_i \,\big], \tag{2}$$

$$h_i^{(\ell+1)} = \sigma\Big(\text{MLP}\big[\, h_i^{(\ell)} \oplus g_i,\ \tfrac{1}{c_i}\sum_{j \to i} \big(h_j^{(\ell)} \oplus g_j\big) \,\big]\Big). \tag{3}$$

The graph representation is the mean of the final node states, passed through an MLP classifier.

3.3 HybridDegradLSTM

HybridDegradLSTM fuses two complementary views of a cascade:

- **LSTM view:** nodes sorted by hop depth form a sequence; a bidirectional LSTM (hidden 128, 2 layers) encodes cascade order, giving $h_{\text{LSTM}} \in \mathbb{R}^{256}$.
- **Graph view:** the DegradGNN branch encodes tree structure, drift and depth, giving $h_{\text{GNN}} \in \mathbb{R}^{128}$.

The concatenation $[h_{\text{LSTM}}, h_{\text{GNN}}]$ is classified by an MLP ($128 \to 64 \to 1$). The model thus captures *both* temporal propagation order and structural/content degradation (Figure 1).

Table 1: UPFD dataset statistics. PolitiFact has only 62 training graphs, making deep models unstable; it is used here solely as evidence for the drift phenomenon.

Dataset	Train	Val	Test
PolitiFact	62	31	221
GossipCop	1,092	546	3,826

Table 2: GossipCop main classification results (3 seeds, mean ± std). HybridDegradLSTM achieves the best result on every metric.

Model	Acc	F1	AUC-ROC	AUC-PR
GCN	0.7823±.0032	.7675	.8686	.8742
GAT	0.9094±.0086	.9099	.9646	.9670
BiGCN	0.9390±.0011	.9388	.9798	.9811
DegradGNN	0.9300±.0037	.9304	.9780	.9790
BaselineLSTM	0.9561±.0028	.9559	.9858	.9859
HybridDegradLSTM	**.9585**±.0030	**.9584**	**.9876**	**.9873**

3.4 Drift-only discriminator

To test whether drift is discriminative *without any learned model*, we extract 12 graph-level statistics per cascade—the mean and standard deviation of each of the five drift dimensions, plus log(size) and max depth—and train a 200-tree random forest. No text features, no graph embeddings, and no neural network are used.

4 Experimental Setup

4.1 Datasets (UPFD)

Features are pre-computed, frozen 768-dimensional BERT embeddings. GossipCop is the primary dataset (Table 1); PolitiFact's 62 training graphs make deep models unstable (Section 6.1).

4.2 Baselines

- GCN, GAT, BiGCN — graph neural networks over the cascade tree.
- BaselineLSTM — bidirectional LSTM over the hop-sorted node sequence.
- BERT-MLP — mean-pooled text features through a 2-layer MLP.

4.3 Hyperparameters

All models: hidden dim 128, two layers, dropout 0.3, AdamW (lr = 10^{-3}, weight decay 10^{-5}), batch size 64 (GossipCop) / 32 (PolitiFact), early stopping patience 8 on validation accuracy, max 20 epochs. Metrics: accuracy, F1, AUC-ROC, AUC-PR. All headline numbers are **3-seed mean ± std**. Hardware: Apple Silicon (MPS), PyTorch + PyG, one seed per process.

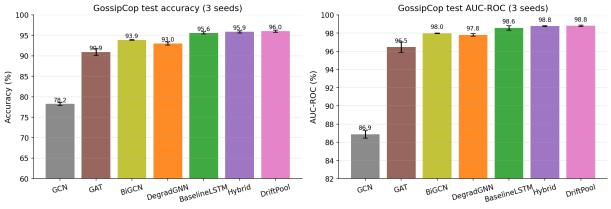


Figure 2: GossipCop test accuracy and AUC-ROC over 3 seeds. Error bars are seed standard deviation.

4.4 Reproducibility note

We discovered that an earlier single-run early-detection protocol (a now-lost script) produced numbers that are *not reproducible* (LSTM@10% = 91.4% vs. 78.7% in our clean re-run). All numbers in this paper come from the saved reproducible scripts. Run-to-run variance on MPS is real even at a fixed seed (∼1–1.5 accuracy points); multi-seed reporting is therefore mandatory.

5 Results

5.1 Main classification results

Table 2 and Figure 2 summarize the GossipCop results. HybridDegradLSTM achieves the best accuracy in our six-model comparison. The margin over the strong LSTM baseline is small (+0.24 acc) and *not statistically significant* at $n = 3$ seeds (paired t-test $p = 0.224$), while the margin over DegradGNN is significant ($p = 0.020$, Cohen's $d = 3.98$); see Section 6.3. This is *not* a state-of-the-art claim: published UPFD results on the same split reach 96.38 (GCNFN) and 97.23 (UPFD) accuracy.

PolitiFact (3 seeds): GCN 0.8311±.0056, GAT 0.7843±.0587, BiGCN 0.7919±.0128, LSTM 0.7330±.1728, DegradGNN 0.4887±.0000, Hybrid 0.7225±.1654. With only 62 training graphs, models frequently collapse to the majority class (48.9%); we do not make model claims on PolitiFact.

5.2 The drift phenomenon: fake news drifts less

We computed per-node drift (Eq. 1, cosine and ℓ_2 components) over the entire test sets—∼220k GossipCop nodes and ∼31k PolitiFact nodes.

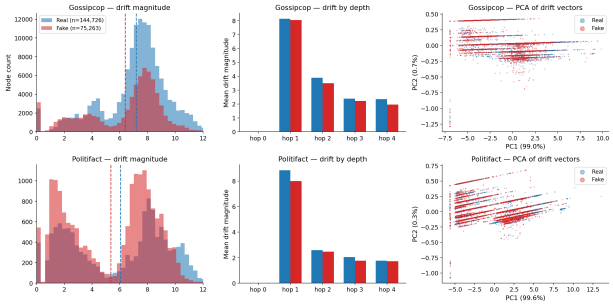


Figure 3: Drift analysis on GossipCop (top) and PolitiFact (bottom): magnitude histograms (left), drift by propagation depth (middle), and PCA of drift vectors (right).

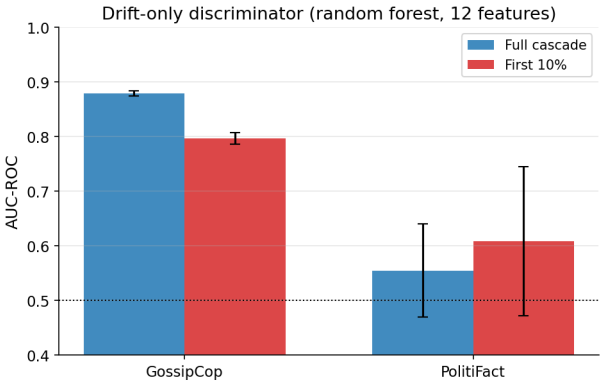


Figure 4: Drift-only discriminator (random forest, 12 features), full cascade vs. first 10%, 5-fold cross-validated.

Table 3: Drift ratio statistics on test nodes. The pooled ratio mixes the content effect with the classes' different propagation-depth compositions; the hop-standardised and cascade-level columns isolate the content effect.

Dataset	Pooled	Hop-std	Casc-lvl	t	p
GossipCop	0.901	**0.977**	0.975	1.97	0.048
PolitiFact	0.883	**0.911**	0.969	0.64	0.52

Table 4: Drift-only discrimination (stratified 5-fold CV on the test split), and its decomposition into content, structural and size feature groups.

Feature set (GossipCop)	Full Acc	Full AUC	10% Acc	10% AUC (all)	10% AUC (no root)
Content drift only (4)	.774	**.846**	.741	.803	.718
Structural only (6)	.772	.853	.747	.796	.686
Size only (2)	.733	.792	.747	.796	.686
All 12 (prev. headline)	.801	.879	.733	.797	.707

The confound. The two classes do not have the same propagation-depth composition: real GossipCop cascades average 75.8 nodes against 39.3 for fake ones, and 80.4% of real nodes sit at hop 1 against 69.0% of fake nodes. Hop-1 nodes carry roughly 2.4× the drift of hop-≥ 2 nodes (8.14 vs. 3.49). The pooled ratio therefore attributes part of a *structural* difference to *content*, and its very large t-statistic is further inflated by the node-count imbalance.

Controlled statistics. Hop-standardisation reweights each hop's per-class mean drift by the combined hop distribution before comparing classes. The cascade-level analysis takes one number per cascade (its mean drift over nodes at depth ≥ 2)—the unit a detector actually sees—and so cannot be inflated by cascades contributing more nodes (Table 3). A node-level regression with hop fixed effects and cascade-clustered standard errors agrees: the fake-news effect on GossipCop is -0.173 (SE 0.034), i.e. -2.4% relative to the real-news mean.

Interpretation. The effect is **real but modest**: roughly 2% on GossipCop, and not significant at cascade level on PolitiFact. Within a fixed depth the direction is consistent (GossipCop hop-2 ratio 0.896, hop-3 0.929, hop-4 0.838) and survives ℓ_2-normalising the BERT embeddings, so it is not an embedding-norm artifact. The per-depth analysis shows the effect emerging at hop ≥ 2; root-node drift is zero by construction for both classes. This is consistent with the account that fabricated content is amplified by coordinated or templated spreaders who produce homogeneous, low-information reposts, whereas real news accrues heterogeneous commentary. We do *not* claim an 11% effect: the pooled figure does not support a causal claim about content, because the classes differ systematically in cascade shape.

5.3 Drift-only discrimination

Using only cascade-level statistics (no text encoder, no neural network), a 200-tree random forest achieves the results in Table 4 and Figure 4.

Decomposition. The headline "12 aggregate statistics" set is not a pure content-drift set: 4 statistics are content drift (cosine and ℓ_2 distance to the parent), 6 are *structural* (depth, child count, rootness) and 2 are pure cascade *size* ($\log n$ and max depth). The content-drift signal is real and the strongest single group, but cascade *size alone* already reaches 0.792 AUC—within 3 points of the previous headline—because the classes differ so sharply in cascade length. The defensible claim is therefore "content drift adds discriminative information on top of cascade shape", not "discriminates without any graph information".

Partial cascades. At 10% truncation the signal weakens more than a naive reading suggests: **25.7%**

Table 5: Does content drift add information beyond cascade shape? Paired bootstrap over identical folds (GossipCop, 1000 resamples).

Feature set	AUC	vs. size (paired bootstrap)
Size only (2)	.792	—
Structural only (6)	.853	—
Content drift only (4)	.846	**+.054** [.040, .068], $p < 10^{-4}$
Size + content (6)	.865	**+.073** [.062, .084], $p < 10^{-4}$
Size + structural (8)	.851	—
All 12	.879	**+.086** [.075, .097], $p < 10^{-4}$

Table 6: Sibling near-duplicate rate. Fake-news cascades contain markedly more near-identical sibling reposts, while overall within-cascade dispersion does not differ.

	Real	Fake	Mann–Whitney p
cosine ≥ 0.90	.9915	.9984	7×10^{-74}
cosine ≥ 0.95	.9266	**.9825**	10^{-287}
cosine ≥ 0.98	.8220	**.8795**	3×10^{-74}
within-cascade dispersion	.9826	.9823	.32

Table 7: Component ablation of DegradGNN (100% data, single-run; reproduced by `scripts/run_ablation_100pct.py`). Removing drift and hop features *improves* accuracy.

Variant	Accuracy	AUC
Full (drift+hop+tree)	0.9247	0.9705
w/o drift	0.9378	0.9829
w/o hop	0.9326	0.9787
w/o drift+hop	0.9365	0.9796
w/o tree (MLP)	0.9634	0.9866
BERT-MLP (text only)	0.9302	0.9751

of GossipCop cascades retain only the root node, whose drift is zero *by construction* rather than by measurement. Restricting to cascades retaining at least one parent–child pair drops the 12-feature AUC from 0.797 to 0.707 (Table 4), and content-only from 0.803 to 0.718.

5.4 Does content drift add information beyond cascade shape?

The decomposition above raises a sharper question than "is drift discriminative?". Since cascade size alone reaches 0.792 AUC, the meaningful question is whether content drift carries information a detector would not already have from the cascade's shape. We test this with a *paired bootstrap* over identical 5-fold out-of-fold predictions (1000 resamples), so the difference between two feature sets is measured on exactly the same data.

Three results matter. First, **content drift significantly beats cascade size alone** (+.054 AUC, $p < 10^{-4}$): the signal is not reducible to "fake cascades are smaller". Second, adding content drift to size is worth more than adding structure to size, so the content channel contributes more than the structural channel. Third, **content drift beats structure at matched size** (+.015, $p = 0.004$): the information is about the *content* relationship between a node and its parent, not about the tree's geometry.

This is the strongest form the central claim can take: *degradation features are not merely a proxy for cascade shape; they carry independent information about how content changes along a propagation tree.*

5.5 Is sibling copying consistent with the effect?

The natural explanation for lower drift in fake cascades is **coordination**: a templated campaign pushes near-identical reposts, so a child's content barely changes from its parent's. Real news accrues heterogeneous commentary, so parent–child distances are larger. This was a narrative in earlier drafts; we test it directly.

For each cascade we measure the **sibling near-duplicate rate**: the fraction of sibling pairs (nodes sharing a parent) whose ℓ_2-normalised BERT embeddings have cosine similarity above a threshold. Sibling similarity is a measurable proxy for repeated content within a local branch; it is not, by itself, evidence that the accounts are coordinated.

If copying is what lowers drift, then cascades with more copying should drift less, regardless of class. Near-duplication correlates negatively with drift ($r = -0.310$, $p = 2 \times 10^{-85}$, $n = 3800$), the relationship holds *within* each class (real $r = -0.320$; fake $r = -0.153$) so it is not a composition effect, and it is unchanged after controlling for cascade size (partial $r = -0.312$). Near-duplication alone discriminates fake from real at 0.831 AUC.

Together, these results are **consistent with** the pathway *content reuse among siblings → smaller parent–child change → lower observed drift.* They do not show that coordination caused the duplication or that duplication mediates the class difference: the embeddings are proxies, and the analysis is observational. This is a plausible mechanism to test with timestamped posts and independently verified coordination labels, not a causal conclusion from the present data.

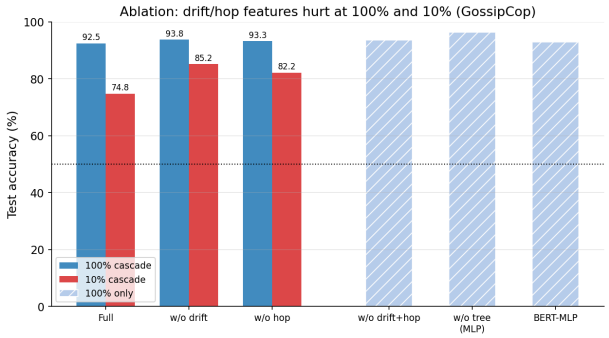


Figure 5: Ablation of drift/hop features in DegradGNN at 100% and 10% cascade sizes (GossipCop).

Table 8: Ablation re-run under partial cascades (seed 1). The ranking is consistent at 10% truncation.

Variant	100% Acc	10% Acc
Full	0.9247	0.7480
w/o drift	0.9365	**0.8521**
w/o hop	0.9354	0.8220

5.6 Ablation study

Finding. Removing drift and hop features improves the model (Tables 7, 8, Figure 5). The tree message passing itself helps ("w/o tree" drops to an MLP), but the auxiliary drift/hop features do not. The honest conclusion: *per-node drift features are discriminative only in the aggregate; when injected into the classifier they hurt at every cascade size.* This negative result reconciles the drift-only discriminator (Section 5.3) and the incremental-value analysis (Section 5.4) with the model-level ablation: the useful information is the *distribution* of drift, not its per-node instantiation.

5.7 Partial-cascade (early detection) analysis

We evaluate models trained on full cascades against truncated test cascades (keeping the shallowest fraction), 3 seeds (Figure 6, Table 9).

All models degrade substantially when only 10% of the cascade is available (from ∼95% to ∼79–81%). Interestingly, the graph models (DegradGNN 81.4%, Hybrid 80.8%) are modestly *more* robust to 10% truncation than the pure LSTM (78.7%)—a +2.5 point mean gap—though per-seed variance overlaps, so this is not significant at $n = 3$. Consistent with the ablation, per-node drift features do not confer truncation robustness. The aggregate drift-only discriminator degrades in relative terms (0.879 → 0.797 AUC), but Section 5.3 shows this still overstates the case: 25.7% of truncated cascades retain only the root, and excluding them drops the signal to 0.707 AUC. Graph-level degradation statistics remain the more promising vector for early detection, but the margin over cascade size alone (0.796 AUC at 10%) is narrow.

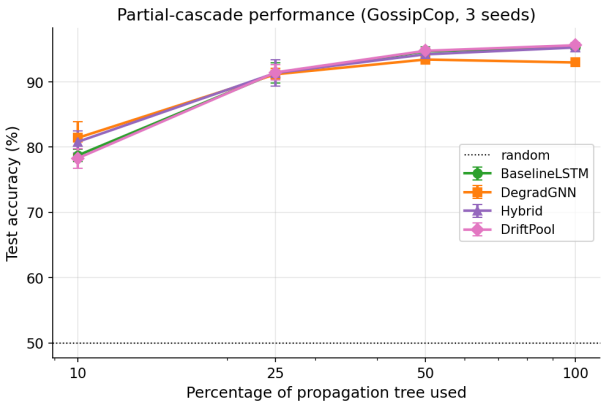


Figure 6: Partial-cascade performance on GossipCop (3 seeds, mean ± std): models trained on full cascades, tested on truncated cascades.

Table 9: Multi-seed truncation results on GossipCop (accuracy, mean ± std). All models degrade substantially at 10%.

Model	10%	25%	50%	100%
BaselineLSTM	.787±.010	.914±.016	.947±.007	.954±.003
DegradGNN	**.814**±.025	.912±.009	.934±.002	.930±.004
HybridDegradLSTM	.808±.017	.914±.020	.942±.010	.953±.005

6 Discussion

6.1 PolitiFact instability

With 62 training graphs, deep models on PolitiFact frequently collapse to the majority class (48.9%), and LSTM/Hybrid show ±17% variance across seeds. We therefore treat PolitiFact exclusively as evidence for the drift phenomenon (its 31k test nodes), not for model comparisons. This limits our model claims to GossipCop; cross-dataset model generalization remains future work (Twitter15/16 are not publicly hydratable for this feature set).

6.2 What we corrected, and why

Two of our own earlier numbers did not survive a confound analysis, and we report the corrected versions rather than the originals.

The pooled drift ratio. An earlier draft led with "fake news drifts ∼11% less, $p < 10^{-70}$". That statistic pools every non-root node across the test set. It correctly describes the node population, but it is *not*

an estimate of a content effect, because the classes have systematically different propagation-depth compositions (Section 5.2). Once the comparison is made at matched depth, the effect is ∼2.3% on GossipCop and not significant at cascade level on PolitiFact. The mechanism we hypothesised—coordinated spreaders producing homogeneous reposts—remains consistent with the controlled numbers, but it is a much weaker claim than "11%".

The "drift-only" discriminator. An earlier draft claimed 0.873 AUC "with no text or graph features". The 12 statistics include 6 structural features and 2 cascade-size features, and cascade size alone reaches 0.792 AUC. Content drift alone reaches 0.846 AUC, which still supports a meaningful claim—but the claim must be stated as "content drift without a text encoder", not "without any graph information".

Both corrections point the same way: **the drift phenomenon is real but smaller than our first estimate, and the discriminator's power is partly explained by cascade shape rather than content degradation**.

6.3 Statistical honesty

- **Drift phenomenon:** the direction is consistent across both datasets and survives depth-standardisation and cascade-level aggregation, but the effect size is ∼2% on GossipCop and not significant at cascade level on PolitiFact. This is our lead claim, stated at its defensible size.
- **Hybrid vs. LSTM:** +0.24% accuracy and +0.19% AUC on average, but $p = 0.224$ / 0.452 at $n = 3$ seeds. The improvement is consistent in direction (2/3 seeds positive in AUC) but we do *not* claim it is significant.
- **Hybrid vs. DegradGNN:** $p = 0.020$, Cohen's $d = 3.98$—the LSTM branch provides a significant gain over the GNN-only model.
- We report mean ± std over seeds and exact per-seed values in Appendix A, and note that MPS training is not bit-reproducible even at a fixed seed.

6.4 Why per-node drift hurts but aggregate drift helps

We hypothesize two mechanisms. (i) **Normalization shift:** drift features are normalized per graph by current maxima; at truncation the depth/children terms shift distribution, systematically misaligning the classifier's input. (ii) **Signal-to-noise:** per-node drift is dominated by idiosyncratic variation; only the aggregate moments (mean, std) separate the classes. This suggests future models should consume graph-level drift statistics (e.g., via set-pooling) rather than concatenating per-node drift.

7 Conclusion

This work makes **content change along propagation edges measurable** and asks whether it reveals anything beyond cascade topology. On GossipCop, fake-labelled cascades exhibit about 2.3% lower drift at matched depth (cascade-level $p = 0.048$); the PolitiFact estimate points in the same direction but is not significant at cascade level ($p = 0.52$). More importantly, in cross-validation on the UPFD test split, content-drift statistics add information beyond cascade size: they reach 0.846 AUC versus 0.792 for size alone, while adding content drift to size outperforms adding structural features (0.865 vs. 0.851 AUC; $p = 0.004$).

Sibling-level repetition provides a plausible explanation for this signal. Fake cascades contain more near-duplicate siblings, and duplication is associated with lower drift within both classes and after controlling for cascade size. Together, these results connect a measurable property of transmission—how much a child changes relative to its parent—to a local pattern of content reuse. They support a copying-based account, while remaining observational rather than causal evidence.

The modeling results yield a practical design principle: **represent drift as a cascade-level distribution, not as an unquestioned per-node feature**. Aggregate summaries carry useful information, whereas injecting drift at each node hurts the tested classifiers. This distinction also sets a boundary for early detection: at 10% cascade size, content-only AUC falls to 0.718 after root-only cascades are excluded, below the 0.796 size-only baseline. The contribution is therefore not a claim that one new detector solves misinformation; it is a measurement framework, evidence that the signal adds information beyond topology, and a concrete rule for how propagation models should consume it.

By shifting attention from **how far a cascade spreads** to **how its content changes as it spreads**, this study opens a complementary direction for propagation-based misinformation research. The next step is to test the same edge-level and cascade-level hypotheses on independent, timestamped datasets, where content reuse and coordination can be measured directly.

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A Per-seed results (GossipCop)

Table 10: Per-seed accuracy and mean AUC-ROC on GossipCop.

Model	Seed 1	Seed 2	Seed 3	Mean	Mean AUC
GCN	.7867	.7804	.7797	.7823	.8686
GAT	.8973	.9151	.9158	.9094	.9646
BiGCN	.9396	.9375	.9399	.9390	.9798
DegradGNN	.9328	.9323	.9247	.9300	.9780
BaselineLSTM	.9530	.9556	.9597	.9561	.9858
Hybrid	.9543	.9608	.9605	.9585	.9876

B Reproducibility index

- `results/multiseed_all.json` — 3-seed main results (both datasets).
- `results/multiseed_driftpool.json` — 3-seed DriftPool results.
- `results/significance.json` — paired t-tests, Cohen's d, drift-ratio statistics.
- `results/significance_5seed.json` — $n = 3$ / $n = 5$ paired statistics.
- `results/drift_standardized.json` — hop-standardised, cascade-level and regression statistics for the drift phenomenon (Section 5.2).
- `results/drift_discriminator_robustness.json` — drift-only RF, full vs. 10%, stratified 5-fold.
- `results/discriminator_decomposition.json` — content vs. structural vs. size decomposition of the discriminator (Section 5.3).
- `results/incremental_value_gossipcop.json` — paired-bootstrap test of whether content drift adds information beyond cascade shape (Section 5.4).
- `results/mechanism_probe_gossipcop.json`, `results/mechanism_linkage_gossipcop.json` — sibling near-duplication rates and their link to drift (Section 5.5).
- `results/ablations.json`, `results/ablations_10pct.json` — ablations at 100% and 10%.
- `results/early_detection_multiseed.json` — multi-seed truncation curves.
- `scripts/` — `make_figures.py`, `analysis_stats.py`, `analysis_drift_standardized.py`, `analysis_discriminator_decomposition.py`, `run_early_detection.py`, `run_ablation_100pct.py`, `run_ablation_10pct.py`, `run_discriminator_robustness.py`.
- `figures/` — all figures embedded above.

All experiments: Apple Silicon (MPS), PyTorch, PyG; one seed per process.